

Regime–Switching Polynomial Diffusions via Topological Hidden Markov Model Inference using Onsager–Machlup Functionals for Asset Pricing

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October 2025

Abstract

This paper develops a unified framework for regime–switching asset pricing models based on polynomial diffusions, hidden Markov models (HMMs), and the Onsager–Machlup (OM) functional. We provide a literature synthesis connecting regime–switching and polynomial diffusion theories in finance, outline a methodology for constructing regime–dependent polynomial diffusion processes, and pose three inferential problems: (i) static parameter estimation via HMM marginal likelihood; (ii) optimal regime path estimation via the Viterbi algorithm; and (iii) most–probable trajectory estimation via OM functionals. These three components are combined to develop a topological HMM framework. We develop an illustrative example for the regime–switching Heston stochastic volatility model under three observation schemes. The paper integrates discrete regime identification and continuous path–likelihood inference, providing a topological stochastic framework for probabilistic path analysis in asset pricing.

1 Introduction

The objective of this work is to build upon existing methods for Hidden Markov Model calibration and estimation for financial asset pricing models. In particular, as part of this Festschrift in Honor of Prof. Robert J. Elliott who made seminal contributions to such methodological and theoretical financial asset pricing models with regime switching, we believe that our methodological contribution will align well with a theme of the edited volume.

We will extend the class of regime switching models in financial applications to the class of polynomial diffusion models and incorporate forward looking derivative market information into the calibration via the Onsager–Machlup (OM) functional. This will combine classical HMM and Viterbi estimation methods, with weighted modifications that utilise the OM functional combined with daily expected move option price intervals, obtained from future market expectations of price range in the weekly options market. This additional information is based on 85% of the value of the at-the-money straddle, which is the conventional industry standard rule of thumb for traders. The range as predicted by the expected move can be used to target high and low prices, and as we will demonstrate, can substantially improve the estimation accuracy of the regime switching models, when incorporated in a topological HMM extension via the OM functional based on the framework proposed in Kashlak et al. [2023].

We will focus the illustrations of this methodology on the stochastic volatility Heston model. This acts as a well established framework for illustration, which is known to have a volatility dynamic that is a special case of the general framework we present our methodology under, namely the polynomial diffusion family of models.

2 Motivation and Background

2.1 Robert Elliott’s Role in Establishing Regime Switching Dynamics in Financial Mathematics

James D. Hamilton [Hamilton, 1989] and Robert J. Elliott (seminal books [Cohen and Elliott, 2015, Elliott et al., 1995, Elliott and Kopp, 2005] and research contributions [Elliott and Krishnamurthy, 2002, Elliott and Osakwe, 2006, Elliott et al., 2010b, Elliott and Siu, 2011, Elliott et al., 2005, Elliott and Siu, 2018]) made complementary but fundamentally distinct contributions to the development of regime-switching models. Hamilton’s seminal work introduced Markov-switching autoregressive models as a practical econometric tool for analysing macroeconomic time series with structural breaks and business-cycle asymmetries. His framework is discrete-time, likelihood-based, and explicitly designed for empirical inference, placing regime switching squarely within mainstream applied econometrics. By providing a tractable filtering and estimation methodology that could be readily implemented, Hamilton transformed regime switching from a theoretical curiosity into a standard empirical modelling paradigm.

In contrast, Elliott’s contributions are rooted in stochastic analysis, filtering theory, and control, and they provide the mathematical infrastructure underpinning regime-switching systems. Elliott formulated regime switching in continuous time using Markov-modulated processes, developed rigorous filtering equations for partially observed regime states, and embedded these models within stochastic control and optimisation problems. This perspective treats regimes not merely as latent indicators for inference, but as unobserved state variables whose conditional distributions evolve dynamically and must be tracked in real time. As a result, Elliott’s framework naturally accommodates

non-linear dynamics, jump processes, and decision-making under regime uncertainty, extending far beyond the econometric scope of Hamilton’s original formulation.

Taken together, Hamilton and Elliott address different but complementary scientific objectives. Hamilton’s work established the empirical relevance and interpretability of regime switching in economic data, while Elliott’s work ensured theoretical coherence, generality, and applicability to continuous-time finance, insurance, and engineering systems. From a modern perspective, many contemporary regime-switching models—particularly those involving filtering, Bayesian learning, or optimal control—implicitly rely on Elliott’s mathematical constructions, even when applied in Hamilton-style econometric settings. Thus, Hamilton provided the empirical entry point for regime switching, whereas Elliott supplied the foundational theory that enables its extension and unification across stochastic modelling disciplines.

In the following sections we will highlight some relevant aspects of these contributions before introducing our proposed novel extensions for this manuscripts contributions.

2.2 Regime-Switching Asset Pricing Models

The incorporation of regime-switching structures into asset pricing models has become a vital approach for capturing the non-linear dynamics and time-varying risk–return relationships observed in financial markets. Traditional linear frameworks, such as the Capital Asset Pricing Model (CAPM) and the Consumption-based CAPM, assume stable parameters over time. However, empirical evidence reveals that market conditions frequently alternate between distinct regimes—transitions between *bull* and *bear* markets characterized by volatility clustering and sudden changes in mean reversion or risk premia—requiring more flexible modelling approaches, see Gordon and St-Amour [2000] and Maheu and McCurdy [2000b].

The seminal work by Hamilton [1989] introduced Markov-switching models, which represent time series dynamics as conditionally dependent on latent Markov states that capture structural shifts between distinct economic regimes. Building on this foundation, Ang and Bekaert [2002], Ang and Timmermann [2012], Edwards [2005], Elliott et al. [2011] and Sarno and Valente [2005] developed versions of regime-switching asset pricing models accommodating shifts in expected returns and volatility, forecasting capabilities and volatility dynamics and spill overs, demonstrating superior explanatory power over static models. Similarly, Guidolin and Timmermann [2008] explored the portfolio implications of regime shifts, showing that accounting for multiple regimes significantly enhances portfolio performance and risk management. In Gouriéroux et al. [2013] the considered regime switching models in fixed income markets. Derivative pricing in the presence of regime switching has been studied in Duan et al. [2002] and Elliott et al. [2005].

Recent developments extend the Markov-switching framework to both discrete-time multifactor models and continuous-time settings. Continuous-time versions embed a finite-state Markov chain into the parameters of stochastic differential equations (SDEs), creating regime-switching diffusions that capture both continuous fluctuations of market factors and abrupt behavioral changes [Elliott

et al., 1995, Hardy, 2001]. For example, Maheu and McCurdy [2000a] integrated regime-dependent volatility dynamics to improve understanding of excess returns predictability, while Hammerschmid and Lohre [2018] analyzed international equity returns under regime shifts. Practical applications include Markov-modulated CIR, Vasicek, and Heston models for interest rates, volatility, and equity returns [Elliott and Siu, 2018, Bulla and Bulla, 2006, Elliott and Siu, 2010, Elliott et al., 2010a, Mitra and Date, 2010, Siu and Elliott, 2022], each demonstrating the framework’s flexibility in capturing asymmetric market behaviours and structural breaks in financial data.

Overall, regime-switching asset pricing models represent a crucial evolution in financial econometrics and financial mathematics, offering a more realistic depiction of market behaviour under uncertainty. As financial markets continue to experience abrupt changes due to macroeconomic and geopolitical shocks, such models remain indispensable for both academic research and practical asset management applications. There is a valuable R package, that was modified for the implementation used in this work, to implement regime switching HMM calibration models Oelschläger et al. [2024]. The modified version of the Topological HMM model estimation via the OM functional is provided in GitHub https://github.com/GTT0N9/Heston_THMM.

2.3 Polynomial Diffusions for Asset Pricing

Polynomial diffusions, formally developed by Cuchiero et al. [2012b] and extended by Filipović and Larsson [2016a], form a tractable class of continuous-time Markov processes that generalize affine processes by allowing both drift and diffusion coefficients to be polynomial functions of state variables. The defining feature of this framework is that the infinitesimal generator maps polynomials of degree $\leq n$ into polynomials of degree $\leq n$, which implies that conditional moments of all orders satisfy closed linear systems. This algebraic property yields closed-form or efficiently computable expressions for moment dynamics and expectations, making polynomial diffusions especially attractive in asset pricing, risk management, and interest rate modelling.

The mathematical foundations of polynomial diffusions rest on the finite-dimensional linear dynamics of coefficients in polynomial moment vectors, enabling eigenfunction expansions and matrix-exponential solutions for pricing operators. This structure allows moments and expectations to evolve tractably, even in high-dimensional settings. Filipović and Larsson [2016a] systematically develops these properties, establishing existence and uniqueness conditions, parameterizations on various state spaces (including cones and compact sets), and methods to construct high-order polynomial models that preserve positivity and other economically important constraints. Important examples include the Ornstein–Uhlenbeck, Cox–Ingersoll–Ross, and Heston models, each belonging to the polynomial class with degree-one or degree-two components.

While polynomial diffusions generalize affine processes—which yield exponential-affine characteristic functions and linear Riccati equations for transform coefficients—the polynomial framework guarantees linear evolution for moments and polynomial expectations. This distinction is significant: polynomial models can capture richer non-linearities while retaining computational tractabil-

ity for moments and certain conditional expectations, making them attractive in applications where full characteristic functions are less convenient or unavailable. The polynomial class has recently gained prominence for its analytic tractability, ability to preserve positivity constraints, and ease of embedding into higher-dimensional volatility or term-structure models. Their combination with regime-switching mechanisms is particularly powerful, allowing one to model both smooth and abrupt structural shifts in financial markets within an analytically tractable framework.

In asset pricing applications, polynomial diffusions have been successfully employed to model state variables driving stochastic discount factors, volatility, and term structure dynamics, see examples in He et al. [2024] and He et al. [2025]. The moment-closure property simplifies pricing of polynomial payoffs and yields closed-form expressions for risk premia approximations. This tractability facilitates estimation via generalized method of moments or likelihood approaches that exploit moment recursions, and permits straightforward incorporation into regime-switching or hidden-Markov frameworks, where polynomial moment dynamics conditional on regimes enable efficient filtering and pricing routines.

Estimation of polynomial diffusion models typically leverages the linear moment equations through simulated method of moments, GMM, and quasi-maximum likelihood approaches that approximate transition densities. For high-dimensional or constrained state spaces, numerical linear algebra techniques—including matrix exponentials and spectral decompositions—become central.

Despite their flexibility and analytic appeal, polynomial models face certain limitations. Capturing heavy tails or extreme nonlinearities may require higher polynomial degrees, which increases parameterization complexity and can complicate interpretation. Linking polynomial specifications to observable economic fundamentals and ensuring identifiability in multivariate settings remain active research areas. Promising directions for extending practical applications include further integration with jump processes, regime switching dynamics, non-Markovian features, and data-driven basis expansions.

One of the contributions we seek to make in this work is to explore advances that utilise the Onsager–Machlup functionals and topological hidden Markov model ideas to enhance inference for polynomial diffusions with complex volatility structures that involve regime switching latent dynamics.

2.4 Econometric Estimation via Hidden Markov Models and Viterbi Algorithm

Hidden Markov Models (HMMs) describe a system where a discrete unobserved Markov chain S_t governs the distribution of observed variables. In finance, HMMs capture hidden economic or volatility regimes influencing observable returns or realized variances. Parameter estimation is typically performed using the Baum–Welch algorithm, an expectation–maximization (EM) method that maximizes the marginal likelihood [Baum et al., 1970].

The Viterbi algorithm [Viterbi, 1967] provides an efficient dynamic-programming procedure to estimate the most probable sequence of hidden regimes, forming a foundation for discrete-time

regime inference in financial econometrics.

Subsequent developments have extended the classical HMM framework to address the complexities of financial data. Elliott et al. [1995] provided a rigorous stochastic filtering formulation for HMMs, establishing a basis for continuous-time and control-based estimation methods. Applications in finance include modelling of regime-switching volatility, credit risk, and portfolio allocation problems where latent states represent underlying market conditions. Extensions to hidden semi-Markov models (HSMMs) allow for non-geometric regime durations, capturing empirically observed persistence in volatility regimes [Bulla and Bulla, 2006]. Moreover, hybrid models combining HMMs with stochastic volatility or diffusion processes have emerged to bridge discrete and continuous frameworks, improving both interpretability and fit to high-frequency data [Hardy, 2001, Filipović and Larsson, 2016b].

2.5 Topological HMM via Onsager–Machlup Functionals in Stochastic Analysis

The Onsager–Machlup (OM) functional, introduced in nonequilibrium thermodynamics by Onsager and Machlup [1953], quantifies the relative likelihood of continuous-time paths generated by stochastic differential equations. For a diffusion

$$dX_t = b(X_t) dt + \sigma(X_t) dW_t, \quad 0 \leq t \leq T, \quad (1)$$

the OM functional assigns to any smooth path $\varphi(t)$ the action

$$I[\varphi] = \frac{1}{2} \int_0^T \frac{|\dot{\varphi}(t) - b(\varphi(t))|^2}{\sigma^2(\varphi(t))} dt + \frac{1}{2} \int_0^T \nabla \cdot b(\varphi(t)) dt. \quad (2)$$

Minimizing $I[\varphi]$ yields the *most probable trajectory*. In this work, we apply OM functionals to polynomial diffusions, showing that they can serve as *path-likelihood functionals* within Hidden Markov frameworks, replacing pointwise emission probabilities with topological path densities, building upon the recent work on topological HMM’s proposed in Kashlak et al. [2023].

The OM functional has deep connections to the theory of large deviations and the pathwise formulation of stochastic processes, as discussed in Freidlin and Wentzell [1998] and Graham and Tél [1985]. It provides a bridge between stochastic dynamics and variational principles, allowing probabilistic models to be interpreted geometrically through their most likely paths or manifolds in state space.

In financial contexts, these ideas underpin likelihood-based inference for diffusion models, particularly where trajectories, rather than discrete observations, drive regime identification or volatility estimation. Cuchiero et al. [2012a] and Filipović and Larsson [2016b] highlight how polynomial and affine diffusions offer analytic tractability for moment and path-based functionals, facilitating OM-inspired likelihood computations.

By integrating the OM formalism with polynomial diffusions, within a Baum-Welch type HMM

estimation and modified Viterbi estimation under the topological HMM framework of Kashlak et al. [2023], in this work we will establish and demonstrate a unifying framework where stochastic volatility, latent regimes, and topological path features can be estimated consistently within Hidden Markov or regime-switching structures.

This synthesis connects thermodynamic principles of path optimality with modern probabilistic modelling in finance, suggesting new directions for state estimation, filtering, and rare-event analysis in continuous-time asset pricing models.

3 Methodology: Regime–Switching Polynomial Diffusion Framework

We consider a continuous–time model in which a finite–state Markov chain $S_t \in \{1, \dots, K\}$ governs the coefficients of a polynomial diffusion process X_t . Conditional on $S_t = i$, the dynamics follow

$$dX_t = \mu_i(X_t) dt + \sigma_i(X_t) dW_t, \quad \mu_i(x) = \sum_{n=0}^N a_{n,i} x^n, \quad \sigma_i^2(x) = \sum_{m=0}^M b_{m,i} x^m. \quad (3)$$

The generator of the regime process is $Q = [q_{ij}]$. When S_t jumps, the coefficients $(a_{n,i}, b_{m,i})$ change instantaneously, leading to a *Regime–Switching Polynomial Diffusion* (RSPD).

This construction yields a class of stochastic processes capable of modelling: continuous diffusive fluctuations within each regime; discrete jumps in dynamics at regime changes; and nonlinear, mean–reverting, and level–dependent volatility effects. Such RSPDs generalize affine regime–switching models while preserving analytic tractability through polynomial moment conditions.

3.1 Example Polynomial Diffusion Models in 1-Dim. and 2-Dim. of Interest

3.1.1 Special Cases: Volatility Models (CIR Process)

For modelling volatility or interest rates, which cannot be negative, a common choice is the **Cox–Ingersoll–Ross (CIR)** process. It is a polynomial diffusion with a linear drift (degree 1) and a square-root diffusion (degree 1).

$$dX_t = \kappa(\theta - X_t)dt + \sigma\sqrt{X_t}dW_t.$$

In the polynomial framework:

- **Drift:** $\kappa\theta - \kappa X_t$ (a linear polynomial).
- **Diffusion:** $\sigma\sqrt{X_t} = \sqrt{\sigma^2 X_t}$ (a linear polynomial under the root).

The key feature of the CIR process is its **level-dependent volatility**: the randomness increases as the level of the process, X_t , increases. This term also ensures that the process remains non-negative

(provided the Feller condition, $2\kappa\theta \geq \sigma^2$, is met), making it an ideal model for variance.

3.1.2 Special Cases: 2-dimensional Heston Stochastic Volatility Model

In the following section we will focus on classes of polynomial diffusion models that are volatility dynamic models. We will extend to a two dimensional class of polynomial diffusion model known as the Heston Volatility model. A multi-dimensional process is classified as a polynomial diffusion if all components of its drift vector and its covariance matrix are polynomials of the state variables. We demonstrate this below.

The Heston model describes the evolution of a 2-dimensional state vector $\mathbf{X}_t = (S_t, v_t)^T$, where S_t is the asset price and v_t is its variance. The system is defined by the following stochastic differential equations (SDEs):

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^1, \quad (4)$$

$$dv_t = \kappa(\theta - v_t)dt + \xi \sqrt{v_t} dW_t^2, \quad (5)$$

where W_t^1 and W_t^2 are standard Wiener processes with correlation ρ , such that $E[dW_t^1 dW_t^2] = \rho dt$. The parameters μ, κ, θ , and ξ are constants.

3.1.3 Checking the Polynomial Conditions

To verify the polynomial property, we must inspect the model's drift vector $\boldsymbol{\mu}(\mathbf{X})$ and its covariance matrix $\boldsymbol{\Sigma}(\mathbf{X})$.

Drift Vector: The drift vector consists of the coefficients of the dt terms for each state variable. For the state vector $\mathbf{X} = (S, v)^T$, the drift is:

$$\boldsymbol{\mu}(S, v) = \begin{pmatrix} \mu S \\ \kappa(\theta - v) \end{pmatrix}$$

We check each component:

- The first component, μS , is a polynomial of degree 1 in S .
- The second component, $\kappa(\theta - v) = \kappa\theta - \kappa v$, is a polynomial of degree 1 in v .

Since all components of the drift vector are polynomials of the state variables, the drift condition is satisfied.

Covariance Matrix: The covariance matrix $\boldsymbol{\Sigma}(S, v)$ is derived from the diffusion (the dW_t) terms. The element Σ_{ij} is the product of the diffusion coefficients of the i -th and j -th processes, multiplied

by their correlation.

$$\begin{aligned}\Sigma_{11} &= (\sqrt{v}S)(\sqrt{v}S) \cdot \rho_{11} = vS^2, \\ \Sigma_{22} &= (\xi\sqrt{v})(\xi\sqrt{v}) \cdot \rho_{22} = \xi^2v, \\ \Sigma_{12} = \Sigma_{21} &= (\sqrt{v}S)(\xi\sqrt{v}) \cdot \rho_{12} = \rho\xi vS.\end{aligned}$$

This gives the matrix:

$$\Sigma(S, v) = \begin{pmatrix} vS^2 & \rho\xi vS \\ \rho\xi vS & \xi^2v. \end{pmatrix}$$

We check each element of the covariance matrix:

- $\Sigma_{11} = vS^2$ is a polynomial in v and S .
- $\Sigma_{12} = \Sigma_{21} = \rho\xi vS$ is a polynomial in v and S .
- $\Sigma_{22} = \xi^2v$ is a polynomial of degree 1 in v .

All elements of the covariance matrix are also polynomials of the state variables. Because all elements of both the drift vector $\mu(S, v)$ and the covariance matrix $\Sigma(S, v)$ are polynomials in the state variables S and v , the Heston stochastic volatility model is, by definition, a **2-dimensional polynomial diffusion model**.

3.2 Constructing a Regime-Switching Polynomial Diffusion Model for Volatility Dynamics

A regime-switching polynomial diffusion model enhances standard volatility models by allowing their parameters to change over time, reflecting structural shifts in market behavior (e.g., moving from a calm to a turbulent state). The model is constructed by combining two components: a continuous-time Markov chain that governs the “regime,” and a polynomial diffusion process whose parameters depend on the current regime.

3.3 The Regime Process (Markov Chain)

The “state” of the market is modeled by a **continuous-time Markov chain**, S_t , which takes values in a finite set of regimes, $\mathcal{S} = \{1, 2, \dots, K\}$.

- Each state $i \in \mathcal{S}$ represents a different market environment (e.g., state 1 = low volatility, state 2 = high volatility).
- The transitions between states are governed by a **generator matrix** (or intensity matrix) $Q = [q_{ij}]$, where q_{ij} is the instantaneous rate of switching from state i to state j .

For a simple two-state model (Calm vs. Turbulent), the generator matrix is:

$$Q = \begin{pmatrix} -q_{12} & q_{12} \\ q_{21} & -q_{21} \end{pmatrix}.$$

Here, $1/q_{12}$ is the average time spent in the calm state before switching to the turbulent state.

3.4 The Volatility Process (Polynomial Diffusion)

Within each regime, the volatility (or variance) process, V_t , follows a 1D polynomial diffusion. The key is that the parameters of this diffusion are now functions of the current state, S_t .

The general SDE for the regime-switching polynomial diffusion is:

$$dV_t = \left(\sum_{k=0}^N a_k(S_t) V_t^k \right) dt + \sqrt{\sum_{j=0}^M b_j(S_t) V_t^j} dW_t,$$

where:

- $a_k(S_t)$ and $b_j(S_t)$ are the polynomial coefficients that now depend on the state of the Markov chain, and
- W_t is a standard Wiener process, which drives the random fluctuations *within* each regime.

When the Markov chain S_t jumps from one state to another, the values of a_k and b_j instantly switch, changing the dynamics of the volatility process.

4 Example: A Two-State Regime-Switching CIR Model

A very common application is to make the parameters of the Cox-Ingersoll-Ross (CIR) model regime-dependent. Let's say we have two regimes: a "calm" state (1) and a "turbulent" state (2).

The volatility process V_t follows:

$$dV_t = \kappa(S_t)(\theta(S_t) - V_t)dt + \sigma(S_t)\sqrt{V_t}dW_t.$$

The dynamics depend on the state S_t :

- **If $S_t = 1$ (Calm Regime):** The parameters are $(\kappa_1, \theta_1, \sigma_1)$. The process reverts to a low long-term mean volatility θ_1 with low volatility of volatility σ_1 :

$$dV_t = \kappa_1(\theta_1 - V_t)dt + \sigma_1\sqrt{V_t}dW_t.$$

- **If $S_t = 2$ (Turbulent Regime):** The parameters switch to $(\kappa_2, \theta_2, \sigma_2)$. The process now

reverts to a higher long-term mean θ_2 with a higher volatility of volatility σ_2 :

$$dV_t = \kappa_2(\theta_2 - V_t)dt + \sigma_2\sqrt{V_t}dW_t.$$

This construction creates a much richer and more realistic model for volatility dynamics, capturing both the continuous fluctuations within a market state and the sudden, structural jumps between different states.

5 Problem Statements: Estimation and Regime Detection

In this section, we outline the main econometric estimation challenges addressed in this manuscript. We first introduce the classical version, before extending to the topological HMM OM functional versions.

5.1 Problem 1: Static Parameter Estimation via HMM Marginal Likelihood

Given discrete observations $\{X_t\}_{t=1}^T$, the marginal likelihood under parameters Θ is

$$p(X_{1:T} | \Theta) = \sum_{S_{1:T}} p(X_{1:T}, S_{1:T} | \Theta), \quad (6)$$

where $S_{1:T}$ denotes the latent regime sequence. Using a discretized Euler–Maruyama approximation for the diffusion transition densities, the Baum–Welch algorithm iteratively maximizes this likelihood, yielding estimates of both the Markov transition matrix Q and the regime–specific polynomial diffusion parameters $(a_{n,i}, b_{m,i})$.

5.2 Problem 2: Optimal Regime Sequence via Viterbi Algorithm

Conditioned on Θ , the most probable sequence of regimes is

$$S_{1:T}^* = \arg \max_{S_{1:T}} p(S_{1:T} | X_{1:T}, \Theta). \quad (7)$$

The Viterbi recursion computes this efficiently via

$$\delta_t(j) = \max_i [\delta_{t-1}(i)q_{ij}] f_j(X_t | X_{t-1}),$$

where $f_j(\cdot)$ is the regime–specific diffusion transition density, followed by backtracking to recover $S_{1:T}^*$, see details in Forney [2005].

5.3 Problem 3: Most Probable Trajectories via the OM Functional

Given $S_{1:T}^*$, the most probable continuous path of X_t is obtained by minimizing the OM functional

$$I[\varphi] = \frac{1}{2} \int_0^T \frac{|\dot{\varphi}(t) - \mu_{S_t^*}(\varphi(t))|^2}{\sigma_{S_t^*}^2(\varphi(t))} dt. \quad (8)$$

The minimizer $\varphi^*(t)$ represents the most likely trajectory conditional on the regime sequence, linking discrete regime inference with continuous path optimization.

6 Solutions via Classical vs Topological HMM Regime Switching

6.1 Classical Hidden Markov Model Solution to Problem 1.

Let $\{S_t\}_{t=1}^T$ be an unobserved regime process taking values in $\mathcal{S} = \{1, \dots, K\}$ and evolving as a homogeneous Markov chain with initial distribution $\eta_j = \mathbb{P}(S_1 = j)$ and transition probabilities

$$a_{ij} = \mathbb{P}(S_{t+1} = j \mid S_t = i), \quad i, j = 1, \dots, K.$$

Let $\{O_t\}_{t=1}^T$ denote the observed process taking values in a space $\mathcal{X}$. Conditional on $S_t = j$, the observation O_t is drawn from an emission density $b_j(\cdot)$ with respect to a dominating measure on $\mathcal{X}$.

The joint likelihood of observations and states is

$$\mathbb{P}(O_{1:T}, S_{1:T}) = \eta_{S_1} b_{S_1}(O_1) \prod_{t=2}^T a_{S_{t-1}S_t} b_{S_t}(O_t).$$

Marginalizing over all possible state sequences yields the observed-data likelihood

$$L(O_{1:T}) = \sum_{S_{1:T} \in \mathcal{S}^T} \eta_{S_1} \prod_{t=2}^T a_{S_{t-1}S_t} \prod_{t=1}^T b_{S_t}(O_t).$$

6.1.1 Baum–Welch Estimation

Define the forward and backward probabilities

$$\alpha_t(j) = \mathbb{P}(O_{1:t}, S_t = j), \quad \beta_t(j) = \mathbb{P}(O_{t+1:T} \mid S_t = j).$$

They satisfy the recursions

$$\alpha_1(j) = \eta_j b_j(O_1), \quad \alpha_t(j) = \sum_{i=1}^p \alpha_{t-1}(i) a_{ij} b_j(O_t),$$

$$\beta_T(j) = 1, \quad \beta_t(j) = \sum_{k=1}^p a_{jk} b_k(O_{t+1}) \beta_{t+1}(k).$$

Posterior state probabilities are

$$\gamma_t(j) = \mathbb{P}(S_t = j \mid O_{1:T}) = \frac{\alpha_t(j) \beta_t(j)}{\sum_{k=1}^p \alpha_t(k) \beta_t(k)}.$$

Transition posteriors are

$$\xi_t(i, j) = \frac{\alpha_t(i) a_{ij} b_j(O_{t+1}) \beta_{t+1}(j)}{\sum_{i', j'} \alpha_t(i') a_{i'j'} b_{j'}(O_{t+1}) \beta_{t+1}(j')}.$$

Parameter updates are given by

$$\hat{\eta}_j = \gamma_1(j), \quad \hat{a}_{ij} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)},$$

with emission parameters updated according to the assumed parametric family (e.g. weighted means for Gaussian emissions).

6.2 Viterbi Regime Decoding Solution to Problem 2

The most likely regime sequence is obtained by the Viterbi algorithm. Define

$$\delta_t(j) = \max_{S_{1:t-1}} \mathbb{P}(S_{1:t-1}, S_t = j, O_{1:t}),$$

with recursion

$$\delta_1(j) = \eta_j b_j(O_1), \quad \delta_t(j) = \max_{i=1, \dots, p} \{\delta_{t-1}(i) a_{ij}\} b_j(O_t).$$

Backtracking yields the optimal state path $\hat{s}_{1:T} = \arg \max_{S_{1:T}} \mathbb{P}(S_{1:T} \mid O_{1:T})$.

6.3 Topological Hidden Markov Model Extension of OM Functional to Problem Statements 1 & 2

In many applications, observations O_t are functions or stochastic processes living in infinite-dimensional spaces where no Lebesgue density exists. Let γ be a Gaussian measure on a locally convex topological vector space $\mathcal{X}$ with Cameron–Martin space $\mathcal{H}(\gamma)$.

In this setting, emission densities $b_j(\cdot)$ do not exist. Instead, each regime j is associated with a mean element $h_j \in \mathcal{H}(\gamma)$, and the emission contribution is defined via the Onsager–Machlup (OM) functional

$$\log b_j(O_t) := \log \left(\lim_{\varepsilon \downarrow 0} \frac{\mathbb{P}(\|O_t - h_j\| < \varepsilon)}{\mathbb{P}(\|W\| < \varepsilon)} \right) = -\frac{1}{2} \|O_t - h_j\|_{\mathcal{H}(\gamma)}^2,$$

up to an additive constant independent of j .

The surrogate log-likelihood becomes

$$\ell_{\text{OM}}(O_{1:T}) = \log \sum_{S_{1:T}} \eta_{S_1} \prod_{t=2}^T a_{S_{t-1}S_t} \exp \left(-\frac{1}{2} \sum_{t=1}^T \|O_t - h_{S_t}\|_{\mathcal{H}(\gamma)}^2 \right).$$

6.3.1 THMM Baum–Welch Updates

The forward–backward recursions are identical to the classical case, with $b_j(O_t)$ replaced by $\exp(-\frac{1}{2}\|O_t - h_j\|_{\mathcal{H}(\gamma)}^2)$. The regime means are updated by

$$\hat{h}_j = \arg \min_{h \in \mathcal{H}(\gamma)} \sum_{t=1}^T \gamma_t(j) \|O_t - h\|_{\mathcal{H}(\gamma)}^2 = \frac{\sum_{t=1}^T \gamma_t(j) O_t}{\sum_{t=1}^T \gamma_t(j)},$$

interpreted as a weighted projection onto the Cameron–Martin space.

Appendix A provides algorithms comparing and constrasting the classical HMM-Baum-Welch-Viterbi solutions against the modified topological HMM extensions.

7 Onsager-Machlup Functional: Definition and Concept

Consider the diffusion process

$$dX_t = b(X_t) dt + \sigma(X_t) dW_t,$$

where $b : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the drift field, $\sigma : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times n}$ the diffusion coefficient, and W_t an n -dimensional Wiener process.

The *Onsager–Machlup functional* provides a formal description of the relative likelihood of sample paths. For a smooth path $\varphi : [0, T] \rightarrow \mathbb{R}^n$, one writes heuristically

$$\mathbb{P}(X_t \text{ stays within a small tube around } \varphi(t)) \approx \exp \left[- \int_0^T L(\varphi(t), \dot{\varphi}(t)) dt \right],$$

as the tube radius $\varepsilon \rightarrow 0$. The quantity $L(x, v)$ plays the role of a “Lagrangian” describing the path density.

In the simplest case of a standard Brownian motion, that is,

$$dX_t = dW_t,$$

the Onsager–Machlup functional is

$$L(\varphi(t), \dot{\varphi}(t)) = \frac{1}{2} |\dot{\varphi}(t)|^2.$$

When drift and space-dependent diffusion are present, more complicated expressions arise, typ-

ically of the form

$$L(x, v) = \frac{1}{2} \|\sigma(x)^{-1}(v - b(x))\|^2 + \frac{1}{2} \operatorname{div} b(x) + (\text{curvature corrections}).$$

These are obtained via small-tube asymptotics, change-of-measure (Girsanov) arguments, or path-integral formalism.

The Onsager–Machlup functional thus encodes the “most probable path” of a diffusion process in analogy with the classical action functional in mechanics.

As such the **Onsager-Machlup functional** is a tool from statistical physics and stochastic calculus that acts as a **probability density for paths**. It assigns a “cost” or “action” to a possible trajectory of a system, allowing one to find the **most probable path** the system will take between two points.

For a system described by a stochastic differential equation (SDE) of the form

$$dX_t = b(X_t)dt + dW_t,$$

the functional $L[\phi]$ for a given smooth path $\phi(t)$ is:

$$L[\phi] = \frac{1}{2} \int_0^T |\dot{\phi}(t) - b(\phi(t))|^2 dt + \frac{1}{2} \int_0^T \nabla \cdot b(\phi(t)) dt.$$

The terms in the functional can be interpreted as follows:

- The first term, $\int |\dot{\phi}(t) - b(\phi(t))|^2 dt$, is the most important part. It measures how much the path’s velocity ($\dot{\phi}$) deviates from the system’s natural drift (b). Paths that “fight” the drift have a high cost.
- The second term, involving the divergence of the drift ($\nabla \cdot b$), is a subtle correction term that arises from the stochastic nature of the problem.

The Onsager-Machlup functional calculates the “effort” for any conceivable path the process could take. A path that cleverly conforms to the drift or trend will have a low effort, while a path that fights directly against drift or trend will have a very high effort. The path that **minimizes** this functional is the most probable path the process will follow.

The Onsager-Machlup functional is used to solve optimization problems for noisy systems. One of its primary applications of relevance to the work in this manuscript involves:

- **Finding the Most Probable Path:** Its main use is to determine the most likely trajectory a stochastic system will follow between an initial and final state. In the context of the Heston regime switching volatility model studied in this work, we will consider smooth paths between the current underlying asset price today and its volatility dynamic and the daily version of the forward projection of price range obtained from the option market implied Expected Move

based on 85% of the value of the at-the-money straddle one week into the future, a choice widely used in industry by derivative traders. The range as predicted by the expected move can be used to target high and low prices, and is especially useful around earnings season. This range can have a smooth path formed from the current dates price at close to this upper and lower future price range via an OM functional, which can then be incorporated into updating information for the regime switching estimation solution via the topological HMM formulation. This should improve the detection time of the regime switching as well as improve the accuracy of the regime detection.

7.1 Remarks on Form of SDE: Ito vs Stratonovich for OM functional

If one derives the path probability density directly from an SDE in the Itô form, the resulting log-probability (or "action") contains a cumbersome correction term. Schematically, it takes the form:

$$I[\phi(t)] \propto \frac{1}{2} \int \frac{(\dot{\phi} - \mu(\phi))^2}{\sigma^2(\phi)} dt - \frac{1}{2} \int \mu'(\phi) dt.$$

The presence of the second term, $-\frac{1}{2} \int \mu'(\phi) dt$, complicates the expression and obscures the analogy to the Lagrangian or the principle of least action found in classical mechanics.

The Stratonovich integral is defined in a way that is more symmetric in time, making it compatible with the ordinary rules of calculus (e.g., the standard chain rule). When the SDE is first converted to the Stratonovich form,

$$dX_t = \mu_s(X_t)dt + \sigma(X_t) \circ dW_t, \quad \text{where} \quad \mu_s(x) = \mu(x) - \frac{1}{2}\sigma(x)\sigma'(x),$$

the path probability density calculation yields a much cleaner result. The Itô-Stratonovich correction term in the drift, $-\frac{1}{2}\sigma(x)\sigma'(x)$, perfectly absorbs the problematic term that arises in the Itô formulation.

The resulting Onsager-Machlup functional is then given by the elegant and intuitive expression:

$$I[\phi(t)] = \frac{1}{2} \int_0^T \frac{(\dot{\phi}(t) - \mu_s(\phi(t)))^2}{\sigma^2(\phi(t))} dt$$

This form is in direct correspondence with a classical Lagrangian, making the interpretation of minimizing the functional as finding the "path of least action" or the "most probable path" straightforward.

In summary, the conversion to the Stratonovich form is not merely a mathematical trick but a strategic choice to work in a framework where the concept of a path's probability is expressed in its most natural form. This simplifies the derivation and yields a final expression for the Onsager-Machlup functional that is both mathematically elegant and conceptually clear.

We will demonstrate this below, where for the homogeneous volatility model, the standard Ito

result is perfectly fine, where as for the inhomogeneous volatility model, it will be advantaegous to conver to the Stratonovich form.

7.2 Functional for a Polynomial Diffusion Model with Time Homogeneous Volatility Function

A one-dimensional polynomial diffusion model is an SDE where the drift function, $b(x)$, is a polynomial of degree N :

$$b(x) = a_N x^N + a_{N-1} x^{N-1} + \dots + a_1 x + a_0.$$

In one dimension, the divergence $\nabla \cdot b(x)$ is simply the derivative $b'(x)$:

$$b'(x) = N a_N x^{N-1} + (N-1) a_{N-1} x^{N-2} + \dots + a_1.$$

Substituting these into the general formula, the Onsager-Machlup functional for a polynomial diffusion model (with diffusion $\sigma = 1$) is

$$L[\phi] = \frac{1}{2} \int_0^T \left| \dot{\phi}(t) - \sum_{k=0}^N a_k \phi(t)^k \right|^2 dt + \frac{1}{2} \int_0^T \left(\sum_{k=1}^N k a_k \phi(t)^{k-1} \right) dt.$$

7.3 Functional for a Polynomial Diffusion Model with Time Inhomogeneous Volatility Function

A 1D polynomial diffusion process, X_t , is defined by the following stochastic differential equation (SDE):

$$dX_t = \left(\sum_{i=0}^N a_i X_t^i \right) dt + \sqrt{\sum_{j=0}^M b_j X_t^j} dW_t,$$

where

- **Drift Term (dt):** The first polynomial, $(\sum a_i X_t^i)$, controls the deterministic tendency of the process. In financial models, this typically creates a **mean-reverting** pull towards a long-term average.
- **Diffusion Term (dW_t):** The second polynomial, under the square root, controls the magnitude of the random fluctuations. This makes the process's volatility dependent on its current level, X_t .

Now to calculate the Onsager-Machlup Functional for a General 1D Polynomial Diffusion we must convert the standard Itô SDE, converts it to the equivalent Stratonovich form, and then applies the standard formula for the OM functional.

7.3.1 The Itô SDE

We consider a 1D polynomial diffusion process, X_t , defined by the following Itô stochastic differential equation (SDE):

$$dX_t = \mu(X_t)dt + \sigma(X_t)dW_t$$

where the drift $\mu(x)$ and squared diffusion $\sigma^2(x)$ are polynomials:

$$\begin{aligned}\mu(x) &= \sum_{i=0}^N a_i x^i \\ \sigma^2(x) &= \sum_{j=0}^M b_j x^j \implies \sigma(x) = \sqrt{\sum_{j=0}^M b_j x^j}\end{aligned}$$

7.3.2 Conversion to the Stratonovich SDE

The probability density of a path is expressed most simply under the Stratonovich interpretation of the SDE. A Stratonovich SDE is written as $dX_t = \mu_s(X_t)dt + \sigma(X_t) \circ dW_t$. The conversion from an Itô drift $\mu(x)$ to a Stratonovich drift $\mu_s(x)$ is given by

$$\mu_s(x) = \mu(x) - \frac{1}{2}\sigma(x)\sigma'(x).$$

This requires calculating the derivative of $\sigma(x)$. Using the chain rule:

$$\begin{aligned}\sigma'(x) &= \frac{d}{dx} \left(\left(\sum_{j=0}^M b_j x^j \right)^{1/2} \right) = \frac{1}{2} \left(\sum_{j=0}^M b_j x^j \right)^{-1/2} \cdot \left(\sum_{j=1}^M j b_j x^{j-1} \right) \\ &= \frac{\sum_{j=1}^M j b_j x^{j-1}}{2\sqrt{\sum_{j=0}^M b_j x^j}} = \frac{\sum_{j=1}^M j b_j x^{j-1}}{2\sigma(x)}.\end{aligned}$$

The Itô-Stratonovich correction term is therefore:

$$\frac{1}{2}\sigma(x)\sigma'(x) = \frac{1}{2}\sigma(x) \left(\frac{\sum_{j=1}^M j b_j x^{j-1}}{2\sigma(x)} \right) = \frac{1}{4} \sum_{j=1}^M j b_j x^{j-1}.$$

Substituting this back, the Stratonovich drift for our polynomial diffusion is:

$$\mu_s(x) = \left(\sum_{i=0}^N a_i x^i \right) - \frac{1}{4} \left(\sum_{j=1}^M j b_j x^{j-1} \right).$$

7.3.3 The Onsager-Machlup Functional Formula

For a process defined by a Stratonovich SDE, the OM functional for a path $\phi(t)$ over the interval $[0, T]$ is given by:

$$I[\phi(t)] = \frac{1}{2} \int_0^T \frac{(\dot{\phi}(t) - \mu_s(\phi(t)))^2}{\sigma^2(\phi(t))} dt,$$

where $\dot{\phi}(t)$ is the time derivative of the path.

7.3.4 OM Functional for Inhomogeneous Volatility SDE

By substituting the specific expressions for $\mu_s(\phi(t))$ and $\sigma^2(\phi(t))$ into the general formula, we arrive at the Onsager-Machlup functional for the specified class of 1D polynomial diffusions:

$$I[\phi(t)] = \frac{1}{2} \int_0^T \frac{\left(\dot{\phi}(t) - \left[\sum_{i=0}^N a_i \phi(t)^i - \frac{1}{4} \sum_{j=1}^M j b_j \phi(t)^{j-1} \right] \right)^2}{\sum_{j=0}^M b_j \phi(t)^j} dt$$

Remark 1 (Justification for the Itô-Stratonovich Conversion in Onsager-Machlup Calculations).

*The conversion from the Itô form to the Stratonovich form of a stochastic differential equation is a crucial and deliberate step in the calculation of the Onsager-Machlup (OM) functional. The fundamental reason for this is that the **probability density of a path** has a much simpler and more intuitive mathematical structure when expressed using the Stratonovich integral.*

The OM functional, $I[\phi(t)]$, is derived from the logarithm of the probability density functional, $P[\phi(t)]$, which quantifies the probability of a process following a specific path $\phi(t)$. This density is formally obtained via a Radon-Nikodym derivative that relates the measure of the diffusion process to that of a standard Wiener process, a result central to Girsanov's theorem. The choice of stochastic integral (Itô vs Stratonovich) directly impacts the form of this derivative.

8 Example Formulation: Regime-Switching Heston Model

The classical Heston model describes asset price S_t and variance v_t by

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^{(1)}, \tag{9}$$

$$dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^{(2)}, \tag{10}$$

where $E(dW_t^{(1)}, dW_t^{(2)}) = \rho dt$. We introduce a Markov chain S_t with K regimes, affecting subsets of model parameters as follows.

8.1 Case 1: Regime Switching in Asset Price Only

Only the drift of the price process depends on regime:

$$dS_t = \mu(S_t)S_t dt + \sqrt{v_t}S_t dW_t^{(1)}.$$

The sequence of regimes is estimated from observed prices via HMM/Viterbi, after which the volatility parameters are estimated conditionally.

8.2 Case 2: Regime Switching in Volatility Only

Here the variance process changes with regime:

$$dv_t = \kappa(S_t)(\theta(S_t) - v_t) dt + \xi(S_t)\sqrt{v_t} dW_t^{(2)}.$$

One first estimates baseline price parameters, then applies Baum–Welch or Viterbi inference on inferred volatility paths to identify volatility regimes.

8.3 Case 3: Regime Switching in Both Asset and Volatility

When both price and realized volatility are observed,

$$dS_t = \mu(S_t)S_t dt + \sqrt{v_t}S_t dW_t^{(1)}, \quad (11)$$

$$dv_t = \kappa(S_t)(\theta(S_t) - v_t) dt + \xi(S_t)\sqrt{v_t} dW_t^{(2)}. \quad (12)$$

Regime inference can proceed jointly (coupled HMM) or separately if $\rho = 0$. The corresponding OM functional for regime i is

$$L_i[\varphi_S, \varphi_v] = \frac{1}{2} \int_0^T \left[\frac{(\dot{\varphi}_S - \mu_i \varphi_S)^2}{v_i \varphi_S^2} + \frac{(\dot{\varphi}_v - \kappa_i(\theta_i - \varphi_v))^2}{\xi_i^2 \varphi_v} \right] dt. \quad (13)$$

Minimization of L_i yields the most probable joint trajectory (S_t, v_t) consistent with the inferred regime sequence.

8.4 Computation of the Onsager–Machlup Action for a Regime-Switching Heston Model

We consider a (regime-dependent) Heston-type stochastic volatility model written in two-factor form for the price level $S_t > 0$ and variance $v_t > 0$:

$$\begin{aligned} dS_t &= \mu_i S_t dt + \sqrt{v_t} S_t dW_t^{(1)}, \\ dv_t &= \kappa_i(\theta_i - v_t) dt + \xi_i \sqrt{v_t} dW_t^{(2)}, \end{aligned} \quad t \in [0, T], \quad (14)$$

where i denotes the regime (state) and $(W^{(1)}, W^{(2)})$ are standard Brownian motions. For simplicity of exposition we treat the driving noises as independent in the action below; a correlation $\rho \neq 0$ is handled by replacing diagonal quadratic forms with the inverse covariance form (see Remark 2).

Theorem 1 (Onsager–Machlup (OM) action and its practical computation). *Let $\varphi_S : [0, T] \rightarrow (0, \infty)$ and $\varphi_v : [0, T] \rightarrow (0, \infty)$ be absolutely continuous candidate paths with a.e. derivatives $\dot{\varphi}_S, \dot{\varphi}_v$. Define the regime- i OM action (negative log-emission up to an additive constant independent of (φ_S, φ_v)) by*

$$L_i[\varphi_S, \varphi_v] = \frac{1}{2} \int_0^T \left[\frac{(\dot{\varphi}_S - \mu_i \varphi_S)^2}{v_i \varphi_S^2} + \frac{(\dot{\varphi}_v - \kappa_i(\theta_i - \varphi_v))^2}{\xi_i^2 \varphi_v} \right] dt. \quad (15)$$

Then $L_i[\varphi_S, \varphi_v]$ can be computed numerically from discretely observed paths $(\varphi_S(t_k), \varphi_v(t_k))_{k=0}^N$ on a grid $0 = t_0 < \dots < t_N = T$ by the Riemann-sum approximation

$$L_i[\varphi_S, \varphi_v] \approx \frac{1}{2} \sum_{k=0}^{N-1} \Delta t_k \left[\frac{(\Delta \varphi_{S,k} / \Delta t_k - \mu_i \varphi_{S,k})^2}{v_i \varphi_{S,k}^2} + \frac{(\Delta \varphi_{v,k} / \Delta t_k - \kappa_i(\theta_i - \varphi_{v,k}))^2}{\xi_i^2 \varphi_{v,k}} \right], \quad (16)$$

where $\Delta t_k = t_{k+1} - t_k$, $\Delta \varphi_{S,k} = \varphi_S(t_{k+1}) - \varphi_S(t_k)$, $\Delta \varphi_{v,k} = \varphi_v(t_{k+1}) - \varphi_v(t_k)$, and $\varphi_{S,k}, \varphi_{v,k}$ are chosen as left endpoints (or midpoints) on each interval.

Proof. The expression (15) is the quadratic “energy” (action) associated with the two-dimensional diffusion (14) under regime i when written in the form

$$dX_t = b_i(X_t) dt + \Sigma_i(X_t) dW_t, \quad X_t = (S_t, v_t)^\top,$$

with drift

$$b_i(S, v) = (\mu_i S, \kappa_i(\theta_i - v))^\top$$

and (diagonal) diffusion matrix

$$\Sigma_i(S, v) = \begin{pmatrix} S\sqrt{v_i} & 0 \\ 0 & \xi_i\sqrt{v} \end{pmatrix}.$$

For an absolutely continuous candidate path $\varphi(t) = (\varphi_S(t), \varphi_v(t))^\top$, the OM/Lagrangian integrand is proportional (up to additive constants not depending on φ) to the squared norm of the “control” term $\dot{\varphi}(t) - b_i(\varphi(t))$ measured in the inverse diffusion metric:

$$\frac{1}{2} (\dot{\varphi}(t) - b_i(\varphi(t)))^\top (\Sigma_i(\varphi(t)) \Sigma_i(\varphi(t))^\top)^{-1} (\dot{\varphi}(t) - b_i(\varphi(t))).$$

Since $\Sigma_i \Sigma_i^\top = \text{diag}(v_i \varphi_S^2, \xi_i^2 \varphi_v)$, its inverse is $\text{diag}((v_i \varphi_S^2)^{-1}, (\xi_i^2 \varphi_v)^{-1})$, which yields exactly the integrand in (15). Finally, given discrete samples on a grid, approximate the time derivatives by forward differences $\dot{\varphi}_S(t_k) \approx \Delta \varphi_{S,k} / \Delta t_k$ and $\dot{\varphi}_v(t_k) \approx \Delta \varphi_{v,k} / \Delta t_k$, and approximate the integral by a Riemann sum, giving (16). This completes the proof. $\square$

Remark 2 (Correlation correction). *If $d\langle W^{(1)}, W^{(2)} \rangle_t = \rho dt$, then the quadratic form in (15) is replaced by the inverse of the instantaneous covariance matrix. Writing the residual vector*

$$r_i(t) = \begin{pmatrix} \dot{\varphi}_S(t) - \mu_i \varphi_S(t) \\ \dot{\varphi}_v(t) - \kappa_i(\theta_i - \varphi_v(t)) \end{pmatrix},$$

the integrand becomes $\frac{1}{2} r_i(t)^\top \Gamma_i(\varphi_S(t), \varphi_v(t))^{-1} r_i(t)$ with

$$\Gamma_i(S, v) = \begin{pmatrix} v_i S^2 & \rho \xi_i S v \\ \rho \xi_i S v & \xi_i^2 v \end{pmatrix}.$$

9 Experimental Case Studies

This section provides controlled experimental case studies to compare classical HMM inference (Baum–Welch estimation and Viterbi decoding) with the proposed topological HMM (THMM) modifications incorporating Onsager–Machlup (OM) based emissions. We focus on a two-regime switching Heston model in which the latent market state alternates between a calm regime and a turbulent regime for the volatility. Using both a baseline and a challenging configuration, we evaluate regime recovery accuracy, parameter estimation stability, and the implications for volatility diagnostics and option-implied quantities.

9.1 Data Simulation

9.1.1 Model set-up: baseline and challenging configurations

We consider a two-regime Heston model with regime process $S_t \in \{1, 2\}$ evolving as a Markov chain with transition matrix

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad a_{ij} = \mathbb{P}(S_{t+1} = j \mid S_t = i), \quad a_{i1} + a_{i2} = 1.$$

In both baseline and challenging configuration, the transition probabilities are specified as

$$a_{11} = a_{22} = 0.1, \quad a_{12} = a_{21} = 0.9.$$

Global simulation settings. Unless stated otherwise, we simulate a two-regime model ($K = 2$) over a one-year horizon ($T = 1$) using $N = 252$ trading days ($\Delta = 1/252$). When intraday sampling is required for realised-volatility construction, we use an intraday grid of size $n_{\text{intraday}} = 2,400$ per trading day. The initial state is fixed at $S_0 = 100$, $v_0 = 0.10$.

Baseline configuration. The baseline setting uses well-separated regimes with clear differences in long-run variance levels and/or volatility-of-volatility, and sufficiently persistent regimes. Provided

parameter values for this configuration are available in Table 1.

Table 1: Baseline simulation parameters (Calm vs. Turbulent).

Regime	μ_i	κ_i	θ_i	ξ_i	ρ	S_0	v_0
Calm ($i = 1$)	0.30	10	0.10	0.05	-0.90	100	0.10
Turbulent ($i = 2$)	0.30	5	0.50	0.05	-0.90	100	0.10

Challenging configuration. In the challenging setting, the two regimes are characterized by similar regime parameters, including comparable long-run variance levels and mean reversion speeds, which makes regime identification more ambiguous. Provided parameter values for this configuration are available in Table 2.

Table 2: Challenging scenario simulation parameters (Calm vs. Turbulent)

Regime	μ_i	κ_i	θ_i	ξ_i	ρ	S_0	v_0
Calm ($i = 1$)	0.50	5	0.10	0.05	-0.90	100	0.10
Turbulent ($i = 2$)	0.50	1	0.20	0.05	-0.90	100	0.10

9.1.2 Simulation procedure and discretization

We simulate daily data over $N = 252$ trading days (one-year horizon) and set the daily time step $\Delta = 1/252$. For each trading day t , conditional on the latent regime $s_t = i$, we generate the daily close-to-close state $(S_{t+\Delta}, v_{t+\Delta})$ using an Euler-type discretization (with standard modifications to preserve positivity of v_t):

$$S_{t+\Delta} = S_t \exp\left((\mu_i - \frac{1}{2}v_t)\Delta + \sqrt{v_t\Delta} Z_t^{(1)}\right), \quad (17)$$

$$v_{t+\Delta} = \max\left\{v_{\min}, v_t + \kappa_i(\theta_i - v_t)\Delta + \xi_i\sqrt{\max(v_t, 0)}\sqrt{\Delta} Z_t^{(2)}\right\}, \quad (18)$$

where $(Z_t^{(1)}, Z_t^{(2)})$ are standard normal with correlation ρ , and $v_{\min} > 0$ is a small numerical floor.

Intraday sampling and realised volatility. To construct a realised-volatility proxy, we additionally simulate an intraday price path within each trading day using $n_{\text{intraday}} = 2400$ evenly spaced points. Specifically, we subdivide the daily interval into $\delta = \Delta/n_{\text{intraday}}$ and evolve the price (and, when required, the variance) on this finer grid using the same discretization logic, producing intraday log-returns $\{r_{t,m}\}_{m=1}^{n_{\text{intraday}}}$. We then compute realised variance/volatility summaries (e.g. $\text{RV}_t = \sum_{m=1}^{n_{\text{intraday}}} r_{t,m}^2$) and a smoothed realised-volatility series $\widetilde{\text{RV}}_t$ via a local smoother. These realised-volatility objects are used as inputs to the HMM/THMM inference and for diagnostics against the true simulated variance process.

Finally, the regime sequence $\{s_t\}$ is generated from the one-step transition matrix A together with an initial distribution η .

9.1.3 Realized volatility, Smoothed realized volatility, and True volatility (latent)

Let $r_t = \log(S_t/S_{t-\Delta})$ denote daily log-returns. We compute a daily realized variance proxy RV_t (e.g., squared returns, or an intraday-based estimator when available such as Parkinson, Garman-Klass or Roger-Satchell based on OHLC data intradaily, see discussions in Alizadeh et al. [2002]). To mitigate measurement noise, we obtain a smoothed series $\widetilde{RV}_t$ using the `lowess` routine in R, which implements LOWESS (locally weighted scatterplot smoothing). LOWESS performs local weighted least-squares regressions of RV_t on time, assigning higher weights to observations closer to t ; the smoothing parameter (span) controls the bias-variance trade-off. We then compare RV_t and $\widetilde{RV}_t$ to the latent instantaneous variance process v_t used in the Heston data-generating process. Plots of the simulated price processes and the simulate volatility, realised volatility and smoother realised volatility under regime switching for the Base Case Model (Table 1) and the Challenging Case Model (Table 2) is provided in Figure 1.

9.1.4 Expected Move for One-Week-Maturity Vanilla Options under the Heston Model

Barchart’s (<https://www.barchart.com/options>) *Expected Move* (EM) summarizes the market-implied price range of an underlying by a given option expiration. Barchart is the leading provider of real-time stock and commodities data. The EM is derived primarily from options-implied information, notably the at-the-money (ATM) straddle and/or the ATM implied volatility (IV), together with the days-to-expiration (DTE). In practice, Barchart often reports the EM as a symmetric interval around the current spot price S_0 ,

$$[S_0 - \text{EM}, S_0 + \text{EM}], \quad (19)$$

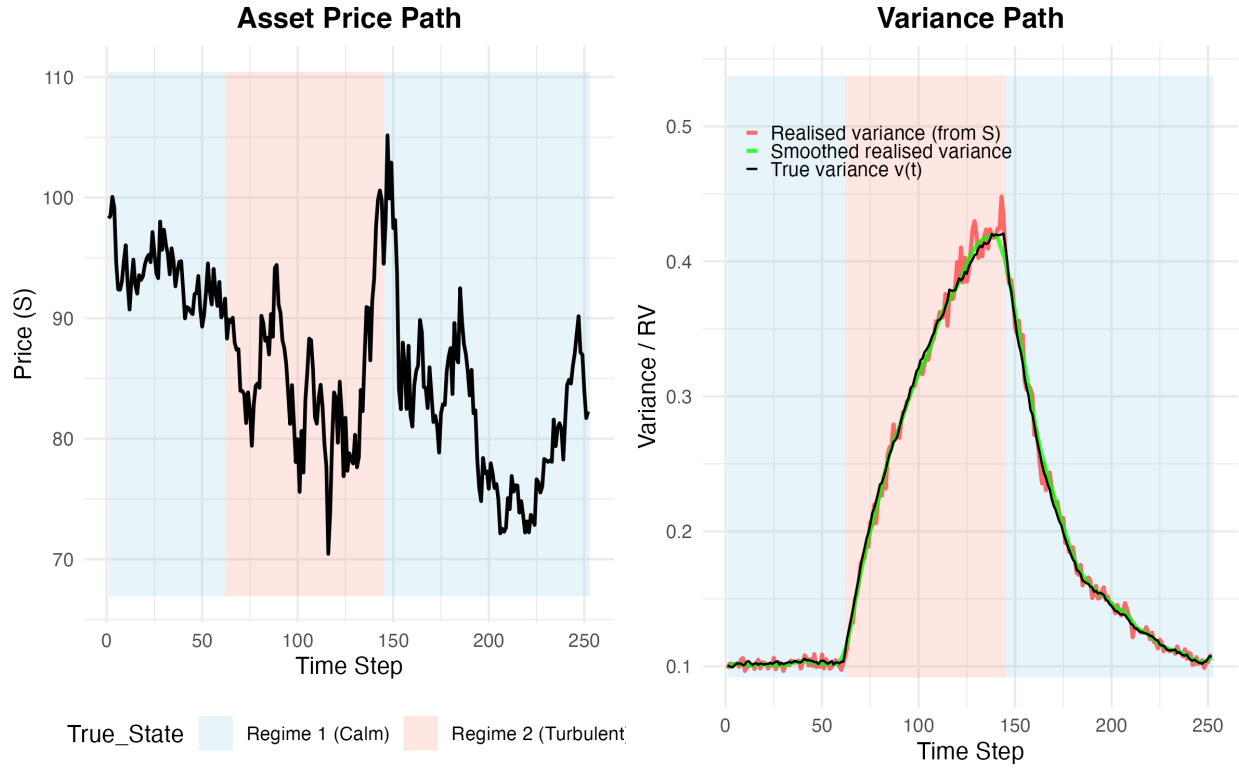
which is frequently used to gauge whether the options market is pricing in a large move around discrete events such as earnings announcements.

ATM-straddle-based approximation. A common implementation is to compute the value of the (near) ATM straddle, i.e., the sum of the ATM call and put premiums at the target expiration,

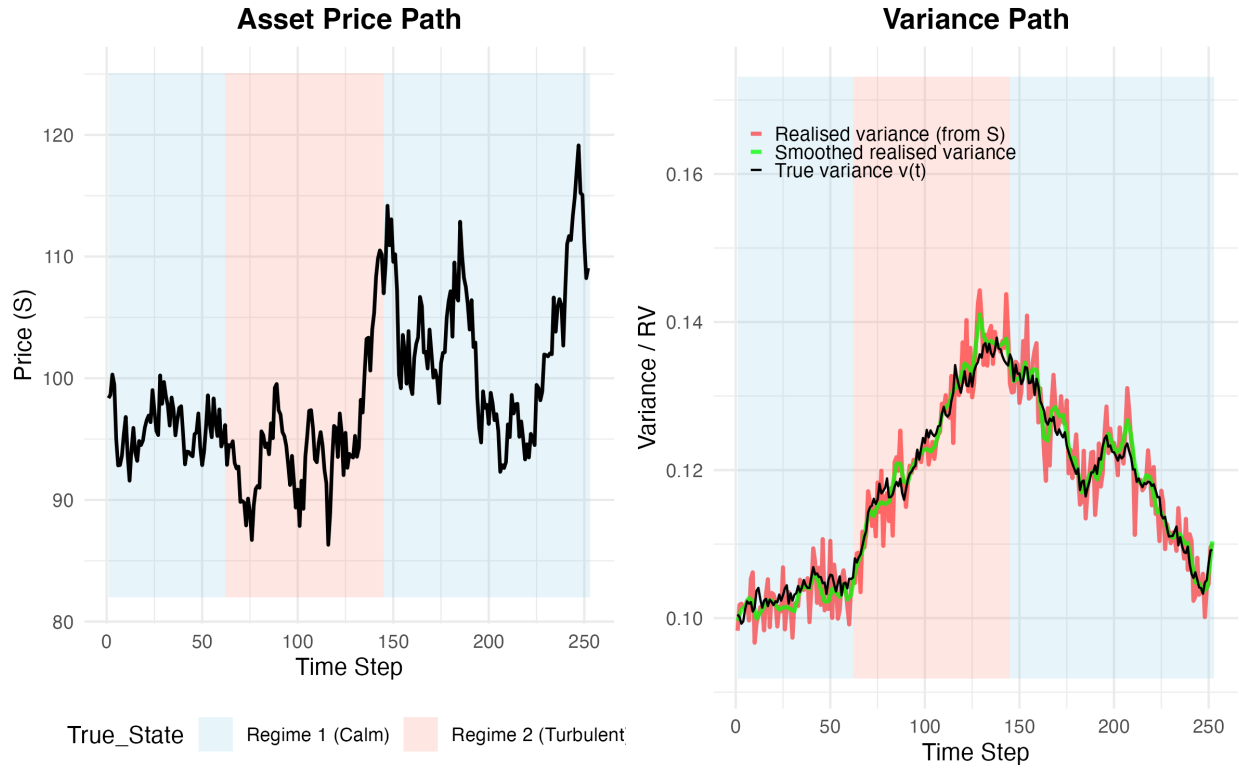
$$\text{Straddle}_{\text{ATM}} := C_{\text{ATM}} + P_{\text{ATM}}, \quad (20)$$

and then apply a scaling factor (often approximately 85%) to obtain the expected move,

$$\text{EM} \approx 0.85 \times \text{Straddle}_{\text{ATM}}. \quad (21)$$



(a) Path Base Case



(b) Path Challenging Case

Figure 1: Simulated paths in two scenarios.

$$C_{\text{BS}}(S_t, K, T) = S_t e^{-qT} \Phi(d_1) - K e^{-rT} \Phi(d_2), \quad (22)$$

$$P_{\text{BS}}(S_t, K, T) = K e^{-rT} \Phi(-d_2) - S_t e^{-qT} \Phi(-d_1), \quad (23)$$

where $\Phi(\cdot)$ is the standard normal distribution and

$$d_1 = \frac{\ln(S_t/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \quad (24)$$

ATM specializations. For *ATM strike* $K = S_t$, and we set $q = 0$

$$C_{\text{BS}}(S_t, K, T) = S_t \Phi(d_1) - S_t e^{-rT} \Phi(d_2), \quad (25)$$

$$P_{\text{BS}}(S_t, K, T) = S_t e^{-rT} \Phi(-d_2) - S_t \Phi(-d_1), \quad (26)$$

and

$$d_1 = \frac{(r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}. \quad (27)$$

Implied-volatility-based approximation. Alternatively, the expected move is often approximated by the one-standard-deviation price change implied by the annualized ATM implied volatility σ_{IV} over a horizon T (in years). Writing

$$T := \frac{\text{DTE}}{252}, \quad \sigma_{\text{IV}} := \frac{\text{IV}}{100},$$

a standard diffusion-based approximation gives

$$\text{EM} \approx S_t \sigma_{\text{IV}} \sqrt{T} = S_t \cdot \frac{\text{IV}}{100} \cdot \sqrt{\frac{\text{DTE}}{252}}. \quad (28)$$

Expected Move–Volatility. In this study, we construct an option-implied expected-move band for volatility that characterizes the range of plausible asset price movements over $[t, t + T]$.

This algorithm determines two strikes (K_t^+, K_t^-) by solving for Black–Scholes Delta levels

$$\Delta^{\text{call}}(S_t, K_t^+, T) = 0.1, \quad \Delta^{\text{put}}(S_t, K_t^-, T) = -0.1,$$

using σ_{IV} as the volatility input. These define an option-implied 80% confidence interval for the asset price over the one-week horizon.

The σ_{IV} : σ_t^+ and σ_t^- can be obtained by solving the following equations:

$$C_t^+ = \text{BSCall}(S_t, K_t^+, H, r; \sigma = \sigma_{\text{IV}}), \quad P_t^- = \text{BSPut}(S_t, K_t^-, H, r; \sigma = \sigma_{\text{IV}}),$$

These implied volatilities are then mapped to volatility endpoints

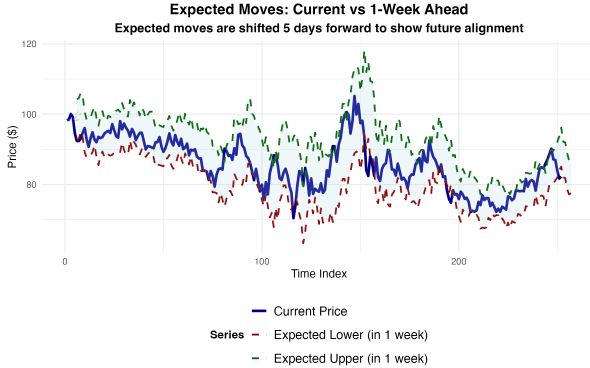
$$\sigma_{t+T}^{\pm} = (\sigma_t^{\pm})^2.$$

We construct linear variance trajectories that connect the current instantaneous variance σ_t^2 to option-implied endpoint variances $(\sigma_{t+T}^{\pm})^2$ over a horizon T . Specifically, for each sign $\pm$, define

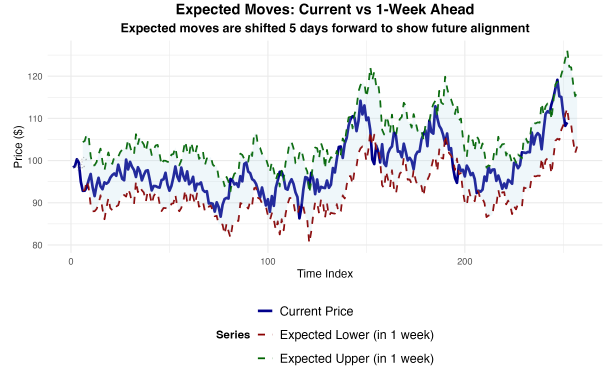
$$\phi_t^{\pm}(\tau) = \sigma_t^2 + \frac{(\sigma_{t+T}^{\pm})^2 - \sigma_t^2}{T} \tau, \quad \tau \in [0, T], \quad (29)$$

so that $\phi_t^{\pm}(0) = \sigma_t^2$ and $\phi_t^{\pm}(T) = (\sigma_{t+T}^{\pm})^2$.

In our simulation studies, we fix a one-week maturity horizon $T = 5/252$. Given the option-implied endpoints $\sigma_{t+T}^{\pm}$, equation (29) yields two candidate option-implied variance paths $\{\phi_t^{\pm}(\tau) : \tau \in [0, T]\}$ over the one-week horizon.



(a) Base case



(b) Challenging case

Figure 2: One-week expected move bands computed from option pricing under the Heston model: base and challenging scenarios.

10 Results and Discussion

We now report inference performance for (i) the baseline benchmark configuration and (ii) the challenging configuration introduced in Section 9. For each configuration, we compare two inferential approaches: (a) classical Hidden Markov Model (HMM) inference based on pointwise emission likelihoods, referred to as the *Standard* approach, and (b) the proposed Topological HMM (THMM) incorporating Onsager–Machlup (OM) functional–based path likelihoods.

Performance is assessed along three complementary dimensions: (i) regime classification accuracy and timing via Viterbi decoding; (ii) segmentation stability, including run-lengths and transition coherence; and (iii) parameter recovery and uncertainty quantification via Baum–Welch estimation and associated confidence intervals. Together, these diagnostics allow us to evaluate not only pointwise accuracy, but also the structural coherence of inferred regime dynamics.

10.1 Baseline Experiment: Standard HMM

10.1.1 Viterbi decoding with true parameters.

Figure 3 reports Viterbi regime decoding for the baseline configuration when true model parameters are assumed known. As expected in this well-separated regime setting, the Standard HMM achieves high classification accuracy, with regime switches largely aligned with the true latent state process. Minor boundary effects are visible around regime transition points, reflecting the inherent lag in discrete-time filtering when regimes change abruptly. Nevertheless, the decoded regime path is stable, with long contiguous runs corresponding to the calm and turbulent regimes.

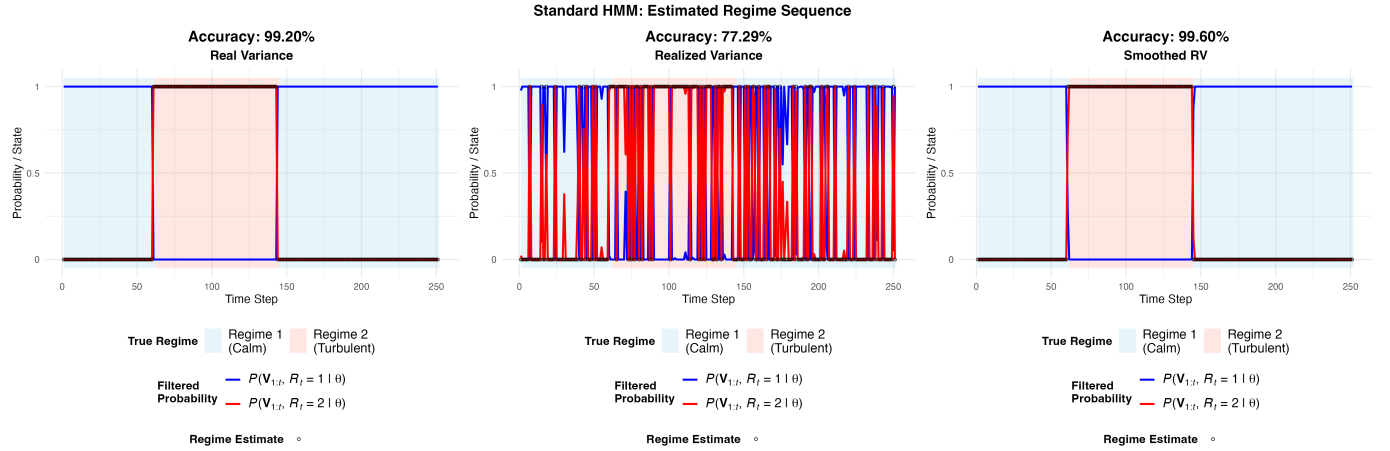


Figure 3: Baseline (Standard): Viterbi decoding with true parameters.

10.1.2 Baum–Welch parameter estimation.

Figure 4 presents the Baum–Welch estimation results for the baseline experiment under the Standard HMM. While the estimated parameters broadly recover the true regime-dependent dynamics, increased variability is observed near regime boundaries. This reflects the sensitivity of pointwise likelihoods to observation noise in realized or smoothed volatility proxies. In particular, regime-specific volatility levels are recovered with reasonable accuracy, but transient misclassification leads to short-lived parameter mixing.

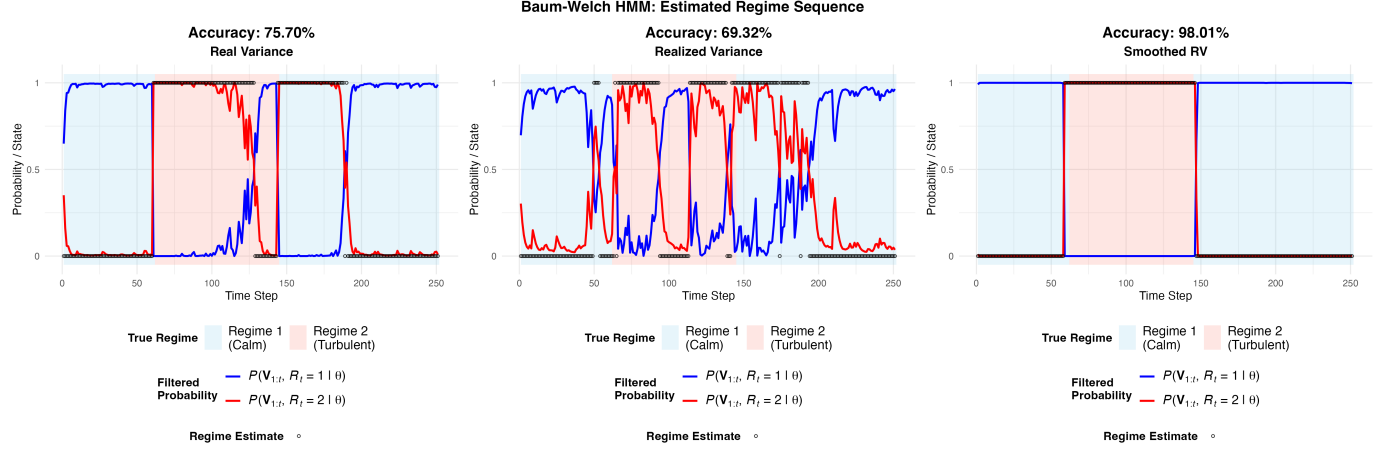


Figure 4: Baseline (Standard): Baum-Welch parameter estimation.

10.1.3 Baum-Welch confidence intervals.

Figure 5 reports confidence intervals for the estimated volatility paths under the Standard approach. Although coverage is satisfactory in aggregate, the intervals widen substantially during transition periods, indicating uncertainty induced by rapid regime switching and observation noise. This behaviour highlights a limitation of pointwise emission models, which do not exploit temporal path structure when assessing likelihood.

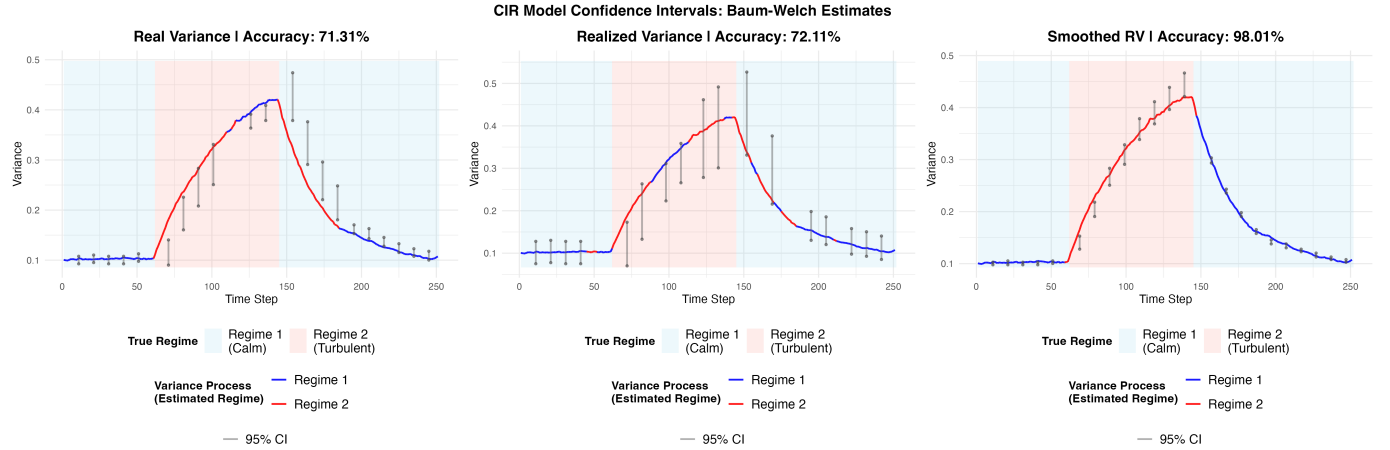


Figure 5: Baseline (Standard): Baum-Welch confidence intervals.

10.2 Baseline Experiment: OM-Functional THMM

10.2.1 Viterbi decoding with true parameters.

Figure 6 shows Viterbi decoding under the OM-functional THMM using true parameters. Relative to the Standard case, regime boundaries are sharper and spurious short-lived regime switches are further suppressed. By incorporating pathwise likelihoods through the OM functional, the THMM

penalizes regime sequences that imply implausible continuous trajectories, thereby enforcing temporal coherence.

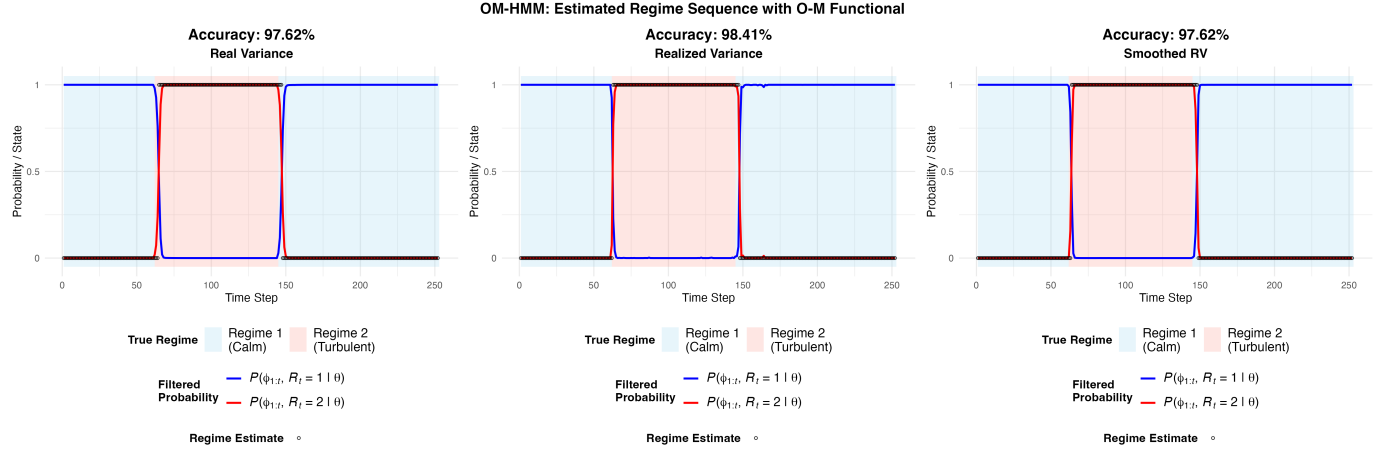


Figure 6: Baseline (OM): Viterbi decoding with true parameters.

10.2.2 Baum–Welch parameter estimation.

Figure 7 illustrates Baum–Welch estimation results under the OM-adjusted framework. Parameter estimates are visibly smoother and more stable across time, particularly for regime-dependent volatility dynamics. The integration of OM-based emissions effectively reduces noise-induced regime oscillations, leading to cleaner separation of calm and turbulent regimes even in the presence of observation error.

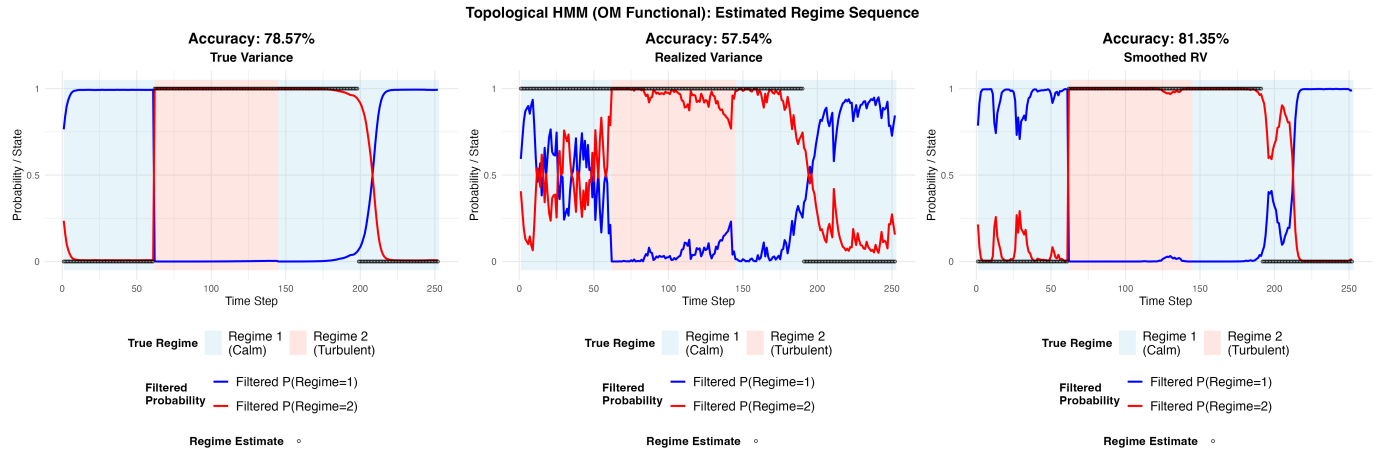


Figure 7: Baseline (OM): Baum–Welch parameter estimation.

10.2.3 Baum–Welch confidence intervals.

Figure 8 shows confidence intervals for the OM-based estimates. Compared to Figure 5, the intervals are tighter and more homogeneous across regimes, reflecting improved inferential efficiency. This

demonstrates that incorporating path information not only enhances regime detection but also yields more informative uncertainty quantification.

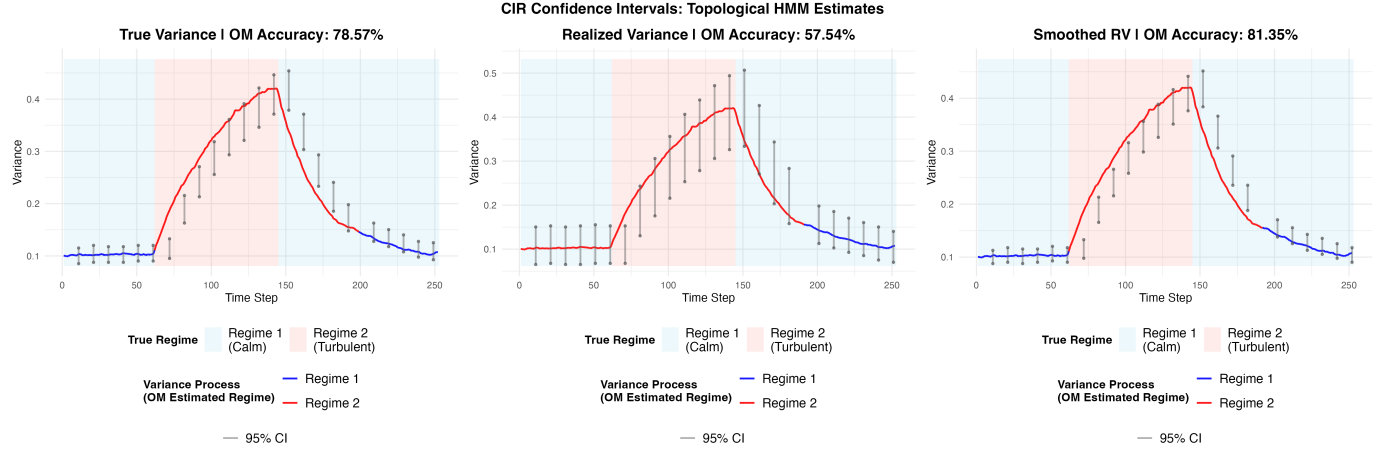


Figure 8: Baseline (OM): Baum-Welch confidence intervals.

10.3 Challenging Case Experiment: Standard HMM

10.3.1 Viterbi decoding with true parameters.

The challenging configuration presents regimes with similar parameter values, making inference substantially more difficult. Figure 9 shows that even with true parameters, the Standard HMM exhibits frequent regime switching and reduced segmentation stability. Regime classification accuracy deteriorates markedly, with short run-lengths and delayed detection of true regime changes.

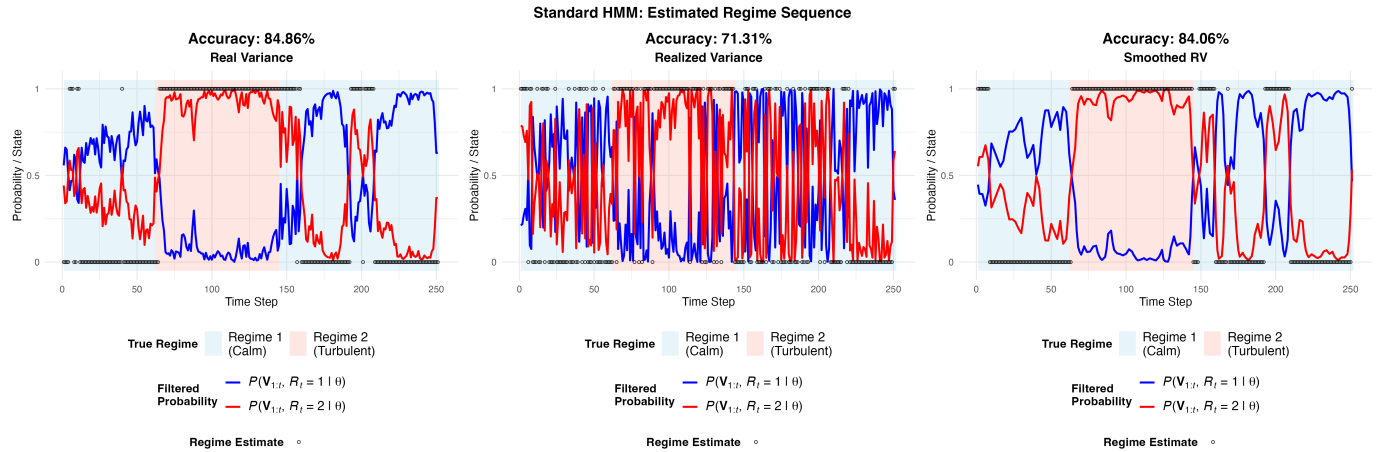


Figure 9: Challenging (Standard): Viterbi decoding with true parameters.

10.3.2 Baum–Welch parameter estimation.

Figure 10 reports Baum–Welch estimates under the Standard model. Parameter recovery is unstable, with substantial mixing between regimes and diminished interpretability of regime-specific dynamics. This highlights the limitations of pointwise emission likelihoods when regimes are weakly separated and observations are noisy.

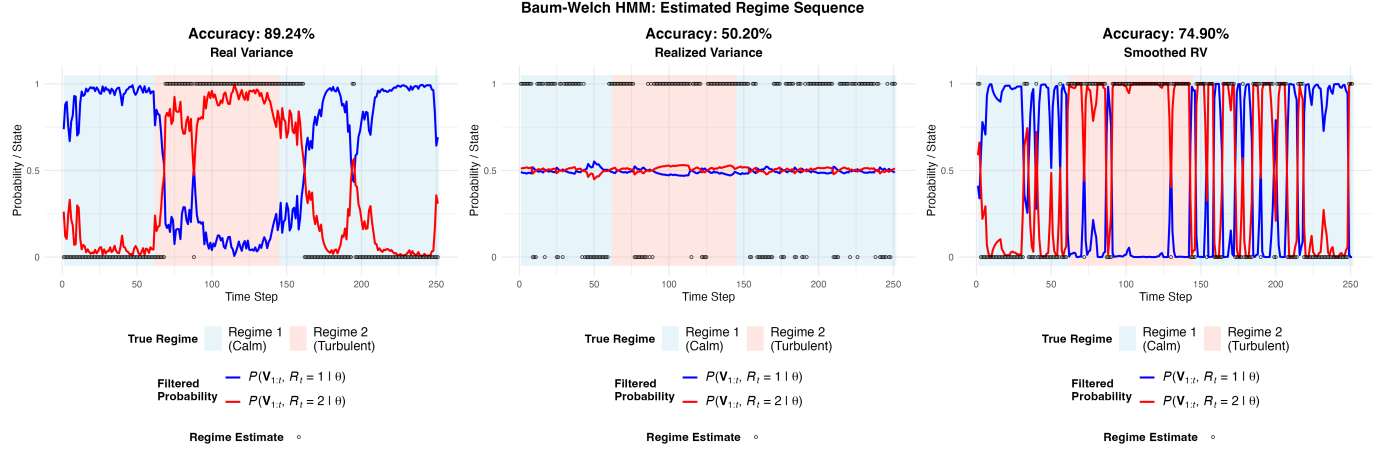


Figure 10: Challenging (Standard): Baum–Welch parameter estimation.

10.3.3 Baum–Welch confidence intervals.

As shown in Figure 11, confidence intervals widen substantially in the challenging case, often overlapping across regimes. This reflects both parameter uncertainty and frequent regime misclassification, reducing the practical usefulness of the inferred model.

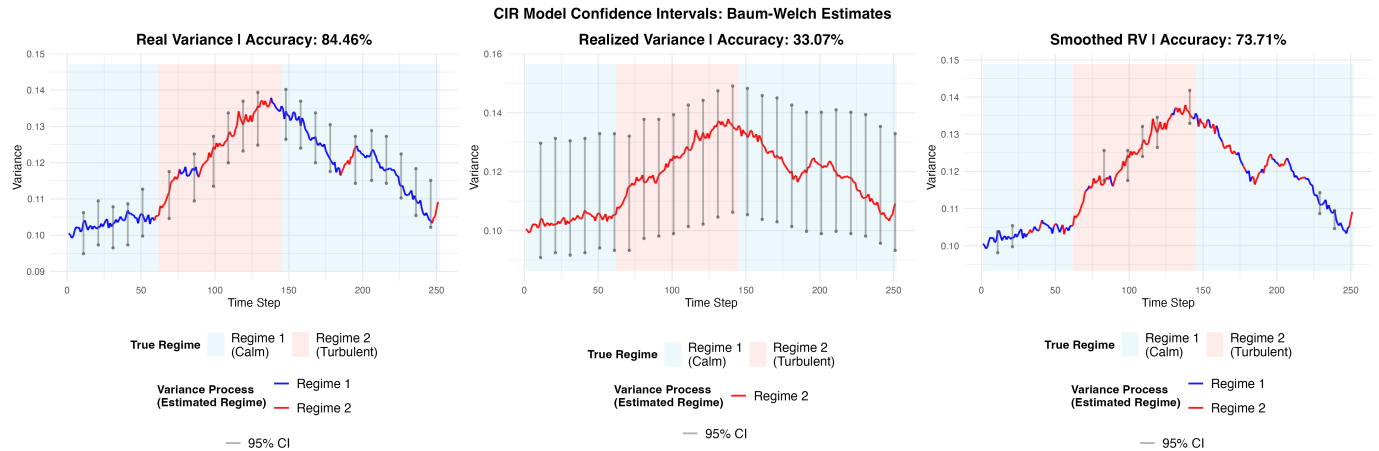


Figure 11: Challenging (Standard): Baum–Welch confidence intervals.

10.4 Challenging Case Experiment: OM-Functional THMM

10.4.1 Viterbi decoding with true parameters.

Figure 12 demonstrates that the OM-functional THMM substantially improves regime decoding in the challenging configuration. Although regimes remain difficult to distinguish, the inferred regime path exhibits longer run-lengths and improved alignment with the true latent states, indicating that pathwise likelihood information provides critical regularization.

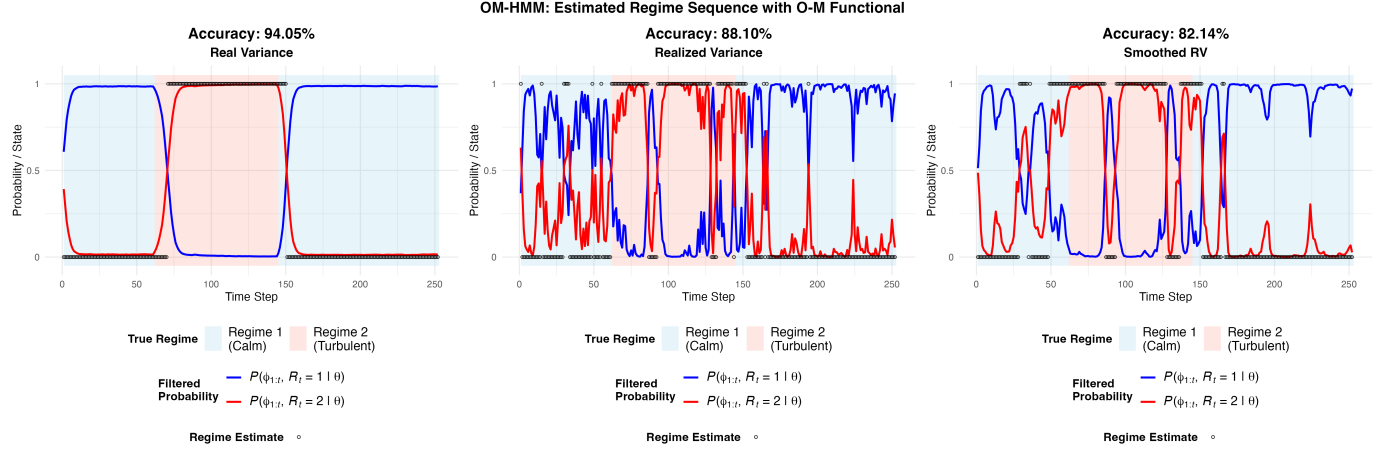


Figure 12: Challenging (OM): Viterbi decoding with true parameters.

10.4.2 Baum–Welch parameter estimation.

Figure 13 shows that OM-adjusted Baum–Welch estimation yields markedly more stable parameter trajectories than the Standard approach. While perfect regime separation is not attainable in this setting, the THMM mitigates excessive regime switching and improves interpretability of the inferred dynamics.

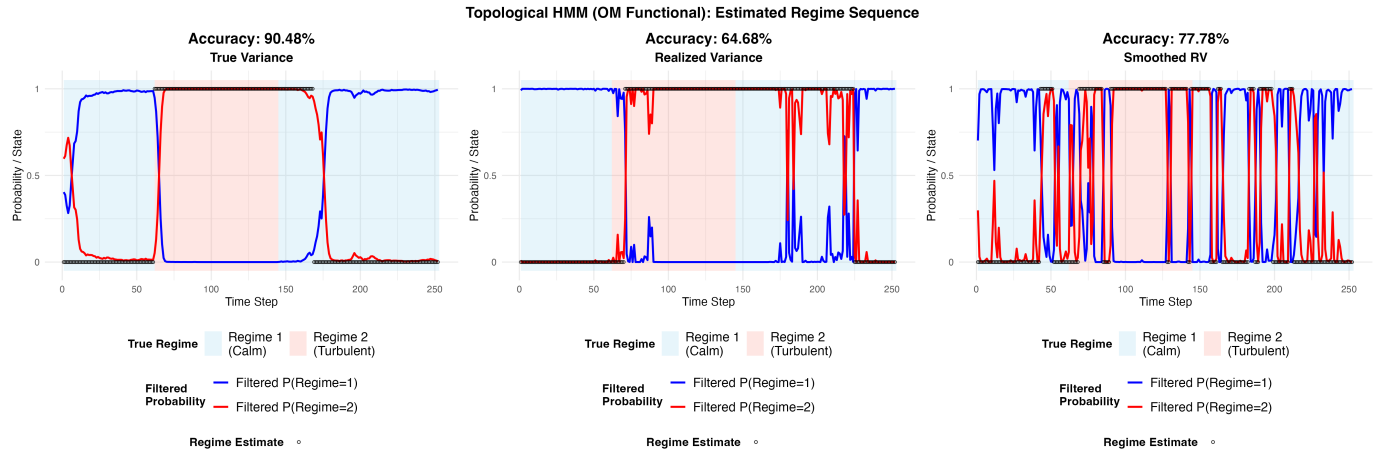


Figure 13: Challenging (OM): Baum–Welch parameter estimation.

10.4.3 Baum–Welch confidence intervals.

Finally, Figure 14 indicates that confidence intervals under the OM-functional framework remain narrower and more structured than their Standard counterparts. This confirms that incorporating OM-based path likelihoods enhances both regime detection and uncertainty quantification, particularly in empirically realistic scenarios where regimes are weakly separated.

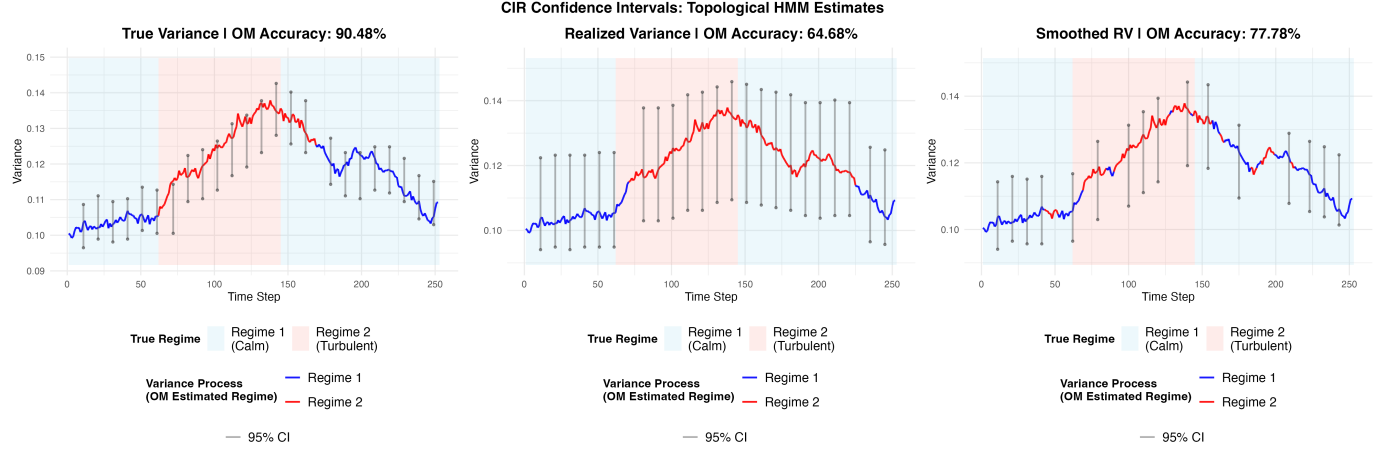


Figure 14: Challenging (OM): Baum–Welch confidence intervals.

11 Conclusion

This paper has developed a unified framework for regime-switching asset pricing models by integrating polynomial diffusion processes, hidden Markov model (HMM) inference, and Onsager–Machlup (OM) path functionals within a topological HMM (THMM) setting. By combining classical Baum–Welch likelihood maximization for static parameter estimation, Viterbi decoding for discrete regime path reconstruction, and OM-based continuous trajectory inference, the proposed methodology bridges discrete latent regime dynamics and continuous-time path likelihoods in a coherent probabilistic framework. This synthesis extends standard regime-switching diffusion models, such as the Heston stochastic volatility model, into a richer topological setting capable of capturing both abrupt structural shifts and smooth most-probable asset and volatility trajectories.

A central contribution of the paper is the demonstration that polynomial diffusions provide a mathematically tractable and flexible foundation for regime-switching dynamics when coupled with OM functionals. The polynomial structure ensures closed-form moment dynamics and facilitates efficient numerical implementation, while the OM functional introduces a principled pathwise likelihood that is particularly well suited to incorporating forward-looking market information. In particular, the use of option-implied expected-move bands as path constraints illustrates how derivative-market information can be embedded directly into regime inference, improving regime detection accuracy and stabilizing parameter estimation in both well-separated and challenging regime configurations.

The numerical experiments highlight the advantages of the proposed THMM/OM framework relative to classical HMM-based inference. Across both baseline and challenging scenarios, OM-adjusted inference improves regime classification accuracy, reduces spurious regime switching, and yields more coherent confidence intervals for latent volatility paths. These results underscore the importance of path-level information in regime identification, especially in settings where regimes are weakly separated or where measurement noise obscures latent state dynamics. From a practical perspective, the framework offers a systematic way to combine realized measures, smoothed volatility proxies, and option-implied expectations within a single inferential architecture.

Several directions for future research naturally emerge from this work. These include extensions to jump–diffusion and Lévy-driven polynomial models, higher-dimensional multivariate systems with regime-dependent correlation structures, and fully Bayesian hierarchical formulations for joint uncertainty quantification over regimes, parameters, and paths. Further exploration of computational strategies for large-scale implementation, including variational and sequential Monte Carlo methods within the THMM/OM framework, also represents a promising avenue. Overall, the proposed methodology provides a flexible and theoretically grounded platform for regime-switching inference in continuous-time asset pricing models, with direct relevance for volatility modelling, derivative pricing, and risk management applications.

12 Contributions

The following outlines contributions of each author:

- GWP: Conceptualization, Formal Analysis, Investigation, Methodology, Validation, Supervision, Writing – Original Draft, Writing – Review & Editing
- MX: Investigation, Methodology, Data, Software, Writing – Original Draft
- ZZ: Investigation, Methodology, Data, Software, Writing – Original Draft
- PVS: Formal Analysis, Validation, Supervision, Writing – Review & Editing

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A Classical vs Topological Algorithms

Algorithm 1 Classical HMM (Baum–Welch + Viterbi)

1. Initialize (η, A, θ_j) .
 2. Iterate:
 - (a) Forward–backward recursion using emission densities $b_j(O_t)$.
 - (b) Update (η, A) via (γ_t, ξ_t) .
 - (c) Update emission parameters θ_j by weighted MLE.
 3. Decode regimes via Viterbi recursion using $b_j(O_t)$.
-

Algorithm 2 Topological HMM (OM-Based Baum–Welch + Viterbi)

1. Choose Gaussian measure γ and Cameron–Martin space $\mathcal{H}(\gamma)$.
 2. Initialize $(\eta, A, h_j \in \mathcal{H}(\gamma))$.
 3. Iterate:
 - (a) Forward–backward recursion using OM emissions $\exp(-\frac{1}{2}\|O_t - h_j\|_{\mathcal{H}(\gamma)}^2)$.
 - (b) Update (η, A) via (γ_t, ξ_t) .
 - (c) Update h_j by weighted Cameron–Martin projection.
 4. Decode regimes via Viterbi recursion using OM emissions.
-

Remark 3. *The classical HMM relies on pointwise likelihoods in finite-dimensional spaces, whereas the topological HMM performs regime inference by comparing entire paths or functions through the geometry induced by the Onsager–Machlup functional.*